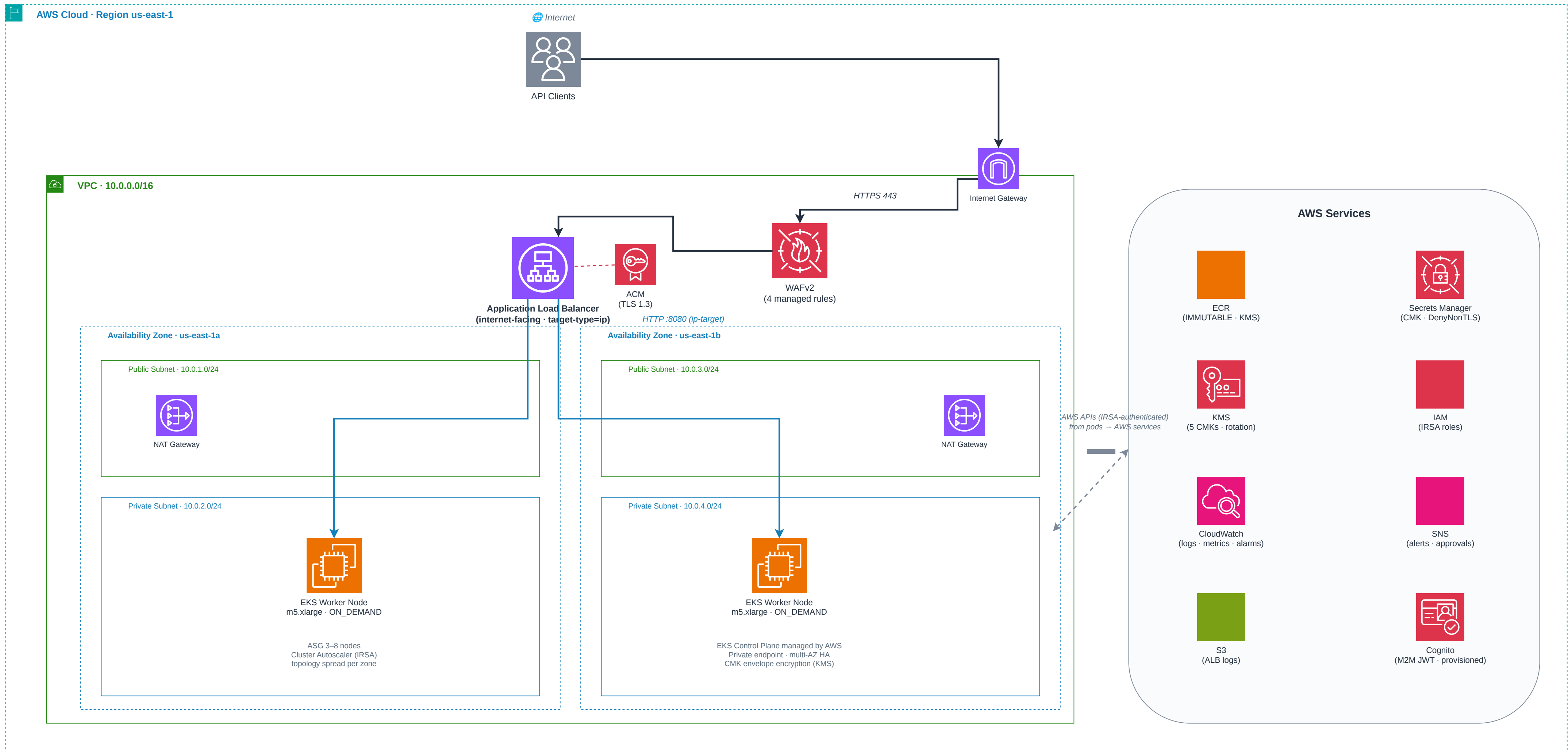


MELI DevOps Challenge — Runtime Architecture (AWS · us-east-1)

Spring Boot 3.5 · Java 21 · EKS 1.31 · 10k RPS target · 99.95% SLA · 2 Availability Zones



EKS Cluster · Kubernetes 1.31 — Logical workloads view

Same manifests applied to nonprod (dev + test) and prod clusters · GitOps via ArgoCD · blue/green via Argo Rollouts

ns: challenge-meli

- Argo Rollout (blue/green)
 - active + preview Services
 - pre: smoke test on /health/ready
 - post: Prom error-rate ≤ 1%
- HPA 3–80 (CPU 60%)
- PodDisruptionBudget · minAvail=1
- ExternalSecret (1h refresh)
- Ingress (alb · ip-target)
- Pod Security: restricted
- readOnlyRootFS · runAsUser=1001

ns: monitoring

- Prometheus (StatefulSet)
 - ServiceMonitor scrapes /metrics
 - 15-day retention
- Grafana (dashboards)
- Loki (logs aggregation)
- Fluent Bit (DaemonSet)
- tails container logs → Loki
- AlertManager
 - error-rate · latency p99
 - pod CrashLoop alerts

Platform controllers (IRSA)

- AWS Load Balancer Controller
 - provisions ALB from Ingress
- Cluster Autoscaler
 - scales ASG 3–8 nodes
- External Secrets Operator
 - syncs Secrets Manager → K8s
- ArgoCD (sync engine)
 - self-heal · prune · 3-min poll
- Argo Rollouts controller
 - blue/green orchestration

EKS managed add-ons

- vpc-cni
 - ENABLE_PREFIX_DELEGATION=true
- coredns (cluster DNS)
- kube-proxy (iptables NAT)
- aws-ebs-csi-driver (IRSA)
 - EBS volumes for Prom/Loki/Grafana

Control plane logs

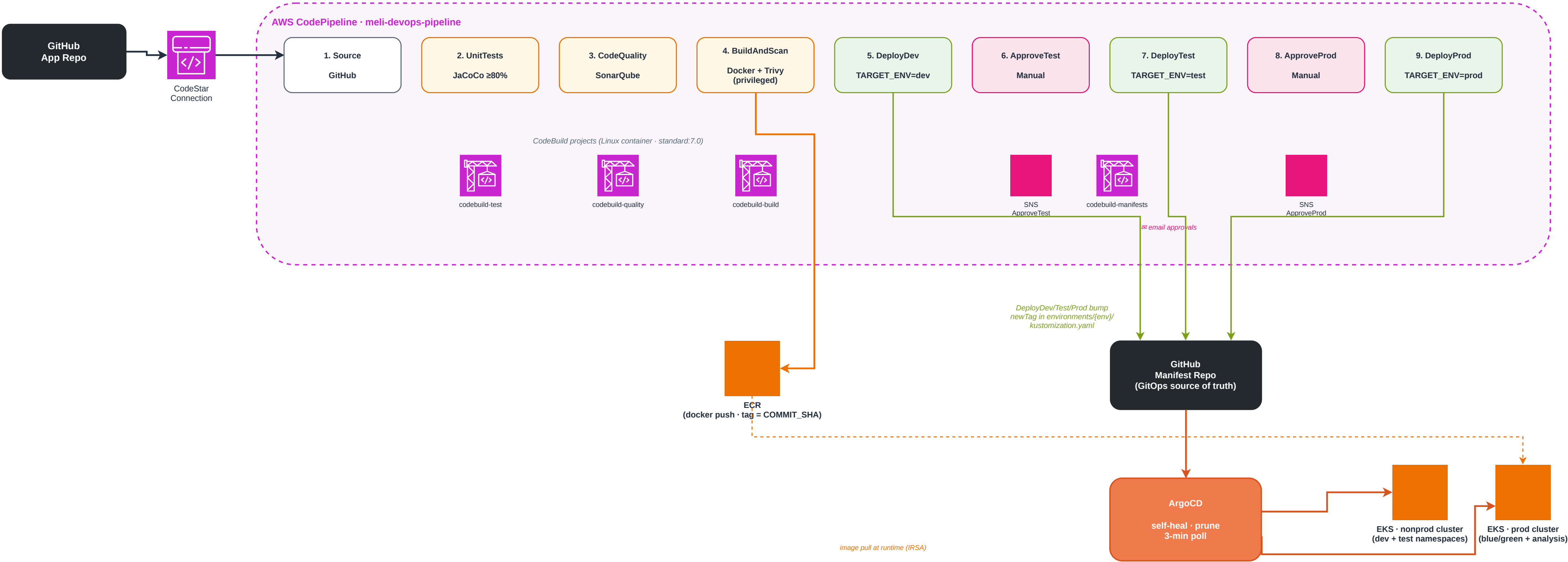
- api · audit · authenticator
- controllerManager · scheduler

Legend

— Request path (HTTPS → HTTP internal) — AWS API call (IRSA) / TLS binding ● Networking ● Security ● Compute ● Storage ● Management

MELI DevOps Challenge — CI/CD & GitOps Pipeline

9-stage CodePipeline → ECR (image) + Manifest Repo (kustomize) → ArgoCD → EKS · blue/green deployment



Argo Rollouts · Blue/Green deployment strategy

Active Service (blue) → receives 100% traffic from ALB
Preview Service (green) → new ReplicaSet, no traffic yet

Pre-promotion analysis — AnalysisTemplate <smoke-test>
GET /health/ready on preview-service · 3 retries · 10s interval

Post-promotion analysis — AnalysisTemplate <error-rate>
Prometheus query: 5xx rate <= 1% over 1-min window · 10 retries

autoPromotion: dev = false (no analysis) · test = true · prod = true
scaleDownDelay: 60s · **abortScaleDown:** 30s · auto-rollback on failure

Pipeline supporting AWS resources

S3
Pipeline Artifacts

KMS
Pipeline CMK

CloudWatch
Build Logs

SSM Parameter
Store (config)

Secrets Manager
(github · sonar)

EventBridge
Pipeline events

IAM
CodeBuild Roles

- S3 + KMS = encrypted artifact store (versioning · 30-day lifecycle)
- CloudWatch = unified log group for all 4 CodeBuild projects
- SSM Parameter Store = pipeline config (MIN_COVERAGE · ECR_REGISTRY · TRIVY_SEVERITY)
- Secrets Manager = github-manifest-token · sonarqube-token (CMK-encrypted)
- EventBridge = pipeline state changes → SNS → email notifications
- IAM = least-privilege roles per CodeBuild project + dedicated CodePipeline role

Legend
— Pipeline flow / writes — Image pull at runtime (IRSA) Stage colors: ● Source · ● Build (CodeBuild) · ● Deploy (manifest update) · ● Approval (Manual + SNS)